



Smishing Detector: A security model to detect smishing through SMS content analysis and URL behavior analysis

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ABSTRACT

Smartphone's popularity and their constant connectivity to the World Wide Web have made these devices vulnerable to phishing and smishing attacks. Phishing is a practice of sending malicious emails to users. Smishing is a combined form of SMS and Phishing in which invaders send SMS containing malicious content to the victim. This content sometimes includes links which redirect the user to websites containing malicious applications and user interfaces. Researchers have proposed various methods in past years to detect smishing but still, we lack a method that significantly avoids false-positive results i.e. falsely categorizing a message as malicious when it is genuine. Hence, we have proposed a model called 'Smishing Detector' to identify smishing messages while reducing false-positive results at every possible step. The proposed method consists of four modules, namely, SMS Content Analyzer, URL Filter, Source Code Analyzer and Apk Download Detector. SMS Content Analyzer analyzes the text message contents. Naive Bayes Classification Algorithm is used to identify the malicious contents and keywords present in the text message. URL Filter inspects the URL to identify malicious features. Source Code Analyzer examines the source code of the website to identify the harmful code embedded in it. Form tag and URL domain present in the source code are also inspected in this module. APK Download Detector identifies whether any malicious file is downloaded while invoking the URL. User consent taken while downloading the file is also inspected in this module. Finally, we have developed a prototype of the proposed system which has been validated with experiments on SMS datasets. In this paper, we have demonstrated the results of each module separately and also we have demonstrated the final results. The results of the experiments show an overall accuracy of 96.29%. We have compared this model with other models proposed by various researchers and we have found that this model covers more security aspects as compared to other models.

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1. Introduction

The development and growth of Information Technology have led us to connectivity through the Internet and increased usage of smartphones. The data shows [1] smartphones are becoming more and more lavish and Smartphone users are accessing the Internet all over the world. Companies like Facebook and Google are striving continuously to make the Internet a more fascinating utility. As technology gets cheaper, the number of smartphones and Internet users will keep on increasing. The rapid growth of Internet users has led to a huge increase in incidents of cybercrimes like phishing and smishing. In 2017, the number of mobile Internet users in India was 351 million. In 2018, the number of users who accessed the Internet through mobile in India was 390 million and is expected to reach almost 469 million

by 2021 [2]. Currently, the Android operating system by Google is the most popular mobile operating system [3], whose market share is greater than 80%. Due to its popularity and lax security, Android-based smartphones are becoming the main target for attackers. User privacy is becoming an important issue in Android devices as users are keeping their crucial information like debit card and credit card details in an Android device. Attackers aim to steal user's private information such as user credentials and financial details by masquerading as a legitimate entity. Mobile device users are more prone to phishing attacks due to the following reasons [4]:

- Users are not able to check the legitimacy of a webpage within the small display of mobile devices because URL addresses are not fully visible within the small display of mobile browsers.
- Mobile users lack awareness about the risks that they might face as a result of adopting less secure behaviors and most

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of the mobile users lack knowledge about security options that can be utilized to avoid phishing attacks.

- Mobile users are in a habit of entering their credentials whenever asked. Several surveys conducted by other authors observed that [5] 40% of Smartphone users enter their passwords into user interfaces at least once. This makes the attacker to come up with fake user interfaces of legitimate websites to obtain the personal and financial credentials of the Smartphone user.

Short Message Service i.e. SMS is very popular among mobile phone users. In India, the usage of smartphones has increased rapidly and people have found text messaging the way to communicate. As text messaging is the most economical option available to reach masses, it is extensively used for offering purchase points of interest and as an advertising medium. Approximately 9132 lac text messages are sent every hour across the world [6]. This increasing use of text messages makes it highly favorable for attackers. More than 90% of SMS messages are opened within 3 seconds [6]. This shows that users feel a sense of urgency which is essential for targeting an attack. SMS phishing is an attack, in which an attacker sends a text message to the user, and that message contains phone numbers or email ids of the attacker or it contains links to malicious web pages, applications or user interfaces that prompt the user to enter their credentials. Users reveal their significant information like user id, password and credit card information through these user interfaces. The attackers acquire user's sensitive information like contact numbers, photos, etc along with monetary gain through this attack. Attackers prefer using SMS instead of email for Phishing [7] because of the following reasons:

- Text messages have higher response rates from users in comparison to emails.
- As sending SMS is an economical option, attackers are able to send SMS in bulk at reduced rates to the victims with a cheap SMS package.
- Users find it difficult to identify whether the URL is a genuine URL or phishing URL by gazing at the link contained in the text messages [7].

SMS Phishing is also known as Smishing which is a compound word of SMS and Phishing. David Rayhawk of McAfee used the word “SMISHING” for the first time on August 25, 2006. Smishing or text message-based phishing involves sending malicious SMS to the user through which the user is prompted to send his sensitive information to smisher. Smisher is an attacker who uses SMS for phishing. Malicious activities of a smishing SMS are categorized in Fig. 1. A Smishing SMS contains any of the following activities [8]:

- **Embedded URL:** In this technique, the attackers send a text message to victims by inserting a URL into it. An example of a Smishing message [9] containing URL is shown in Fig. 2(a). Once the user reads the messages and click the URL, the following activities are initiated:
 - The malicious code embedded in the URL installs into the victim's phone.
 - Some URL redirects the user to a malicious phishing website which looks similar to a legitimate website and prompts the user to fill user credentials or credit card details in the form provided.
 - *.apk file is downloaded into the victim's phone which turns out to be malware and it causes malicious activities later on.

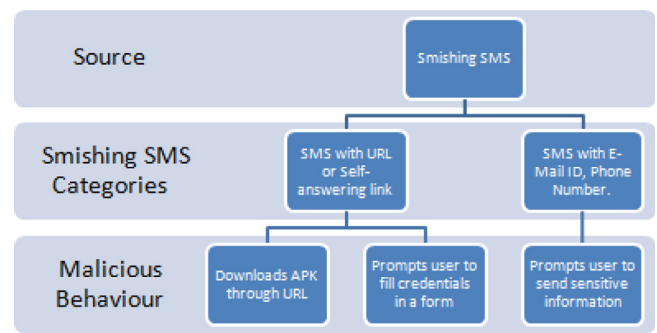


Fig. 1. Malicious activities of a smishing SMS.

- **Phone number and/or Email address:** The attackers sends an SMS to the victim declaring free coupons, discounts or free gifts along with a phone number or email id mentioned in it so that the victims contact the attacker on the phone number or email address and gradually attackers request their credentials. Example of smishing messages [9] containing phone number and email id is shown in Figs. 2(b) and 2(c).
- **Self-Answering SMS:** The attackers send a self-answering message to the victim asking the user to Subscribe or Unsubscribe a service. These links redirect the user to malicious websites.

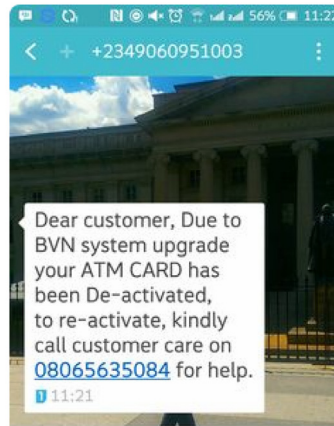
During smishing, attackers design the user interface wisely so that the user is unable to identify the minor differences between a legitimate website and fake website created by the attacker. They copy the source code of the legitimate website to create a fake web page that will look similar to a legitimate web page but these websites could re-direct the user to other malicious links. Also, attackers modify the URL to create a fake URL that looks like URL of a legitimate website. The attacker misplaces a character or misspells the URL to redirect it to a malicious website or file.

The literature survey reveals that there are a few strategies proposed by previous researchers to distinguish smishing SMS from legitimate SMS. While analyzing the existing methods, we found that most of them used the blacklist approach [10], whitelist approach and heuristic approaches [11,12] to detect smishing attacks. However, the blacklist method is not efficient because URL domains of smishing messages are changed fast (age of the domain is very less) and they use URL shortcut service (short URL). So attackers keep on changing the domain name frequently for phishing attacks which makes the blacklisting technique ineffective. Some researchers have used a heuristic approaches but features of an SMS content alone cannot determine it is smishing or not. The presence of some keywords and URL in the message may indicate a smishing but it might be a legitimate message sent by an online shopping website or mobile service provider. It can be a spam message but it cannot be categorized as a smishing message until and unless we find that the URL is redirecting the user to a malicious website trying to steal the credentials of the user or it is downloading a malicious file automatically. If we classify the spam message as a smishing message, it indicates a false-positive result. False-positive results can be defined as incorrectly labeling a legitimate SMS as smishing SMS. It is essential to check the authenticity of the URL contained in a message before categorizing it as smishing [8]. The existing methods proposed by other authors for detection of smishing messages have some limitations. So, it is essential to come up with a new and efficacious method that identifies smishing messages avoiding false-positive results.

Hence, we proposed a model called Smishing Detector based on SMS content Analysis and URL behavior analysis that has low



(a) Smishing using URL



(b) Smishing using Phone Number



(c) Smishing using the Email ID

Fig. 2. Examples of Smishing messages.

false-positive results. We have done the content analysis of the smishing SMS and have also analyzed the behavior of the URL included in it. We have analyzed the source code of the URL to check its malicious behavior and re-directions.

What we propose in this paper:

- A method that can detect Phishing initiated through text messaging.
- An embedded APK Download Detector module which verifies the behavior of the URL included in the SMS.
- A Source Code Analyzer module that inspects the legitimacy of the Source code and its re-directions.
- A user interface which helps the user to select or skip the various steps involved in detecting the Smishing SMS in case if the user is already aware of the legitimacy of the text message.
- To experiment with a real-time application of the system, a prototype of the system is developed and the results of the model are verified using SMS datasets.

The remaining part of this paper is arranged as follows: A summary of the research work done by other authors in this field is discussed in Section 2. Our proposed system is presented in Section 3. The architecture and flowchart of the model are depicted in Sections 3.1 and 3.2 respectively. Section 4 explains the results of the experiments conducted on the prototype of the proposed system. A comparison of the suggested system with other related works is presented in Section 5. The Conclusion of the proposed work is given in Section 6.

2. Related work

In recent years, researchers are focusing more on smishing due to its popularity in mobile attacks. Some researchers have proposed models to detect smishing messages [10,11,13–17]. Some of the spam detection models which used similar approaches are also discussed here [18,19]. Many researchers have discussed smishing strategies and smishing detection approaches to bring awareness among users and researchers [20–23]. This section gives an overview of various research work presented by other authors in the context of Smishing. As smishing is a category of phishing and similar techniques are used for phishing detection, we have discussed some phishing detection approaches [12,24–28].

In a smishing study proposed by Joo et al. [13], the author suggested a system called ‘S-Detector’ for identifying smishing

messages. S-Detector contains four components, namely, SMS monitor, SMS analyzer, SMS determinant, and Database. This model inspects the URL to analyze whether .apk file is downloaded or not. If an executable file is downloaded, the SMS is categorized as smishing. They have used the Naive Bayes classification algorithm to analyze the contents of the messages. Another study proposed by [14] suggested a rule-based method for identifying Smishing messages. They have identified nine rules to filter smishing messages from legitimate messages. Further, these rules were trained using different classification algorithms like Decision Tree, RIPPER, and PRISM. The performance evaluation of the approach shows more than 99% true negative rate. A research work proposed by Sonowal et al. [11] proposed a model named SmiDCA for detecting smishing messages using a machine learning approach. The proposed model extracted 39 features from smishing messages using correlation algorithm. Machine Learning algorithms were used and experiments were conducted on different datasets. Experimental evaluation of SmiDCA displayed an accuracy of 96.40% using Random Forest classifier. Goel et al. [10] proposed a smishing detection system called ‘smishing classifier’ for identifying Smishing messages. This system inspects the contents of the SMS and SMS keywords using Naïve Bayesian Classifier. The proposed framework verifies the existence of URL in SMS and it also inspects the mobile number of the sender. Further, login page appearance and download of APK file are also evaluated in this model. A. Kang et al. [15] discussed various types of phishing and smishing attacks. They have proposed a URL validation test to check the authenticity of the URL included in the message. They have also discussed the smishing box approach as a security measure against downloaded applications during a smishing attack. In another work proposed, the author [16] used a content-based approach to detect smishing messages. A machine learning algorithm is used for identifying the most frequently used keywords in smishing SMS. Further, the appearance of the login page and downloading of the .apk file is also evaluated in this model to inspect the maliciousness of the URL. A recent work proposed by Ankit et al. [17] proposed a feature-based model for detecting smishing messages using a machine learning approach. The proposed model extracted some features of smishing messages and analyzed the results using 5 Machine Learning algorithms on the same dataset. Experimental evaluation of this model displayed an accuracy of 98.74%.

Some authors have proposed spam SMS detection model using similar approaches. In a study presented by Yadav et al. [18], the author presented ‘SMSAssassin’ a spam SMS detection model

using the Naive Bayes algorithm and Support Vector Machine (SVM). Authors have used crowd sourcing to keep the spam database updated. Users having SMSAssassin application on their mobile device can use the spam messages identified on their device to update the Spam Keyword Frequency list to enhance the performance of the system. The authors have also presented [19] the design and implementation of SMSAssassin which performs machine learning-based SMS content filtering to distinguish spam messages. They have designed and implemented an Android application which categorizes and handles different kind of messages which makes the SMS management simple for the user. This application also offers an interface to select the notification preference to the user.

Some researchers have discussed phishing and smishing detection approaches to bring awareness among users and researchers. Foozy et al. [20] proposed taxonomy of phishing detection on a mobile device. They have stated and elaborated various phishing techniques like Bluetooth phishing, SMS phishing, voice phishing, and mobile application phishing. The researchers also compared and evaluated various phishing detection techniques. Hossain et al. [21] elaborated various types of phishing attacks. They have also discussed various phishing mitigation techniques and best policies which should be followed to prevent phishing attacks on mobile devices. Their work is aimed at bringing more security awareness to mobile users. Diksha et al. [22] presented a detailed study of phishing attacks. They have discussed various mobile phishing attacks conducted by attackers and their countermeasures. They have also discussed the taxonomy of phishing solutions, techniques used by authors, various challenges involved in the detection of phishing attacks and they have also elaborated various phishing datasets. In a recent work, Sandhya et al. [23] discussed various smishing attacks performed by attackers. Author has also suggested some policies which can be adopted by the user to deal with the smishing attacks. Various techniques and methodologies used for mitigating smishing attacks are also discussed in this paper.

In a study to detect phishing websites, the author [24] proposed a model called 'PhiDMA', incorporating five layers. They have also implemented a sample of this system which offers an interface for visually impaired persons. Experimental results have shown an accuracy of 92.72%. Wu et al. [25] proposed a phishing detection technique called "MobiFish" that detects phishing attacks on mobile devices. For mobile applications, they have designed "AppFish" and for mobile web pages, they have designed "WebFish". The proposed method examines the URL, the IP address of the URL and HTML source code of the website to detect the maliciousness of the webpage. Zhang et al. [26] proposed a system that uses the features of phishing URLs, such as hosted features and lexical features to inspect the URL. They have used a machine learning classifier to detect the phishing sites on the basis of the selected features. They have shown high accuracy of their proposed system by experimenting it on multiple datasets. Mohammad et al. [27] suggested a phishing website classification system. They have investigated 17 efficient features for the detection of phishing websites and developed a new rule for each feature. They have considered the frequency of each feature in their datasets to identify the most popular feature in the detection of phishing websites.

A novel system proposed in [28], called CANTINA which is a content analysis method to detect phishing websites using TF-IDF measures. TF-IDF score is calculated for each word in a document and then five words with the highest values are selected. A lexical signature is found on the basis of selected words, which is fed into a search engine. If the domain name of N top search results matches the domain name of the current web page, authors declared it as legitimate website else phishing website. Sophie

et al. [12] suggested a heuristic approach for detecting phishing websites. Based on the characteristics of phishing URLs and web pages, they have defined 20 heuristics tests. They have also developed an active anti-phishing toolbar called Phishark. From the experimental results, they have shown that the combination of URL-based features and HTML-based features is very effective in distinguishing legitimate websites from phishing websites.

The existing methods proposed by other researchers used rule-based approaches [10,13,14,16] and heuristics approaches [11,12] to detect smishing attacks. Rule-based approaches included some rules which covered the flow of smishing attacks and its re-directions. But rule-based approaches do not cover the different versatilities of the smishing attacks. Heuristic approaches analyze the contents of the text message. Applying machine learning algorithms on SMS contents only does not detect the versatile smishing trends. We strongly feel that it is necessary to analyze the behavior of the URL link present in the text message for the detection of smishing.

3. Proposed work

3.1. Architecture of the proposed system

The proposed model is a Smishing detector which is comprised of 4 modules, namely; SMS Content Analyzer, URL Filter, Source Code Analyzer and Apk Download Detector as depicted in Fig. 3.

SMS Content Analyzer verifies the presence of URL, self answering link (SAL), phone number and email id in the SMS. Messages containing URL or SAL are transferred to URL Filter. Messages containing email id and phone number are processed for blacklist check. Then messages are forwarded for text pre-processing. Text pre-processing is conducted to convert the text into a form that can be used for text analysis. Text pre-processing includes removing all punctuations and special strings, converting each word to lower case, splitting words to tokenize, stemming i.e. converting each word to its root form and preparing a word vector corpus. Then we categorize the messages on the basis of keywords present in it. Keywords contained in the message are classified using TfidfVectorizer and Naive Bayes classifier. TfidfVectorizer converts each word in the message to the feature index in the feature vector matrix. In each vector, the numbers represents the TF-IDF score of each word selected as a feature. These feature vectors are used as input to Naive Bayes Classifier. Naive Bayes Classifier predicts the result based on the learning from the feature vector matrix.

URL Filter first converts the short URL to Long URL. Then, it looks for the URL in Blacklist. URL found in the blacklist is categorized as Smishing. In the next step, URL Filter verifies four features of the URL, namely, the age of domain, presence of @tag, presence of hyphen and number of dots present in the URL to check the authenticity of the URL. If the threshold of the above features is greater than or equal to three i.e. 75% or above, then we categorize the message as smishing. If the threshold is less than three, we pass on the URL to Source Code Analyzer.

Source Code Analyzer verifies the presence of any form tag in the source code. If form tag is present in the source code, Source Code Analyzer compares the domain of the request URL in source code with the domain of the actual URL invoked. If the domain is different, the message is classified as Smishing. On the other hand, if the domain is same, the URL is transferred to APK Download Detector.

APK Download Detector checks for any file downloading while invoking the URL. This checking is actually carried on by the proposed system without visiting the website. The base name of actual URL is extracted to assess whether it contains .apk extension as part of the base name. This process does not involve invoking the URL or downloading the APK file. It also

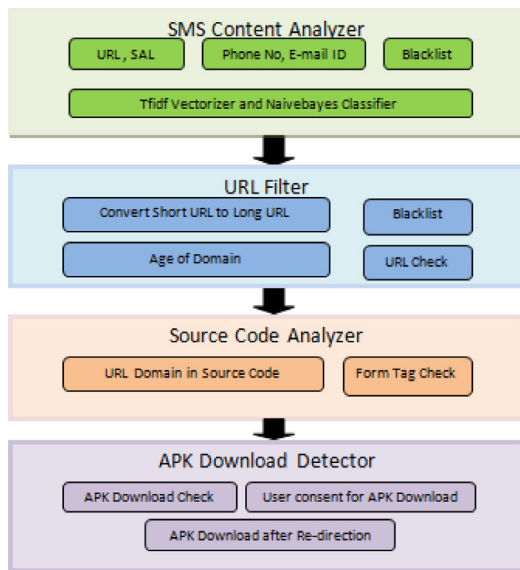


Fig. 3. Architecture of the Proposed System.

checks whether the file is downloaded with user permission or not. Sometimes files are downloaded after re-direction of web pages on phishing websites. Our model also checks for APK downloading after re-direction of the web pages.

3.2. Flowchart of the proposed system

In this section, we elaborate on the flow of the proposed system. The flowchart of SMS Content Analyzer is shown in Fig. 4, URL Filter in Fig. 5, then Source Code Analyzer in Fig. 6, and finally APK Download Detector is depicted in Fig. 7.

3.2.1. SMS Content Analyzer

SMS Content Analyzer analyzes the contents of the message as depicted in Fig. 4. The entire working scenario of SMS Content Analyzer is drafted in the following steps:

Step 1(a): When an SMS is received, it is passed on to SMS Content Analyzer (Fig. 4). Here, the SMS content is analyzed and the system verifies the existence of URL or SAL (Self Answering Link) in the message. If URL or SAL is present in the SMS, then the message is processed to URL Filter (Step-2(a)).

Step 1(b): If URL or SAL does not exist in the SMS, we check for the presence of email id or phone number in it. If the phone number or email id is not present in the message, it is classified as legitimate message.

Step 1(c): If the phone number or email id is present in the SMS, then system checks for email id and phone number contained in the message in the corresponding blacklists. If phone number or email id contained in the message does exist in the blacklist, message is categorized as smishing.

Step 1(d): If email id or phone number is not present in the blacklist, then the system analyzes the keywords present in the message to classify the SMS using the machine learning algorithm.

Step 1(e): If the prediction of the algorithm is malicious based on the learning, message is classified as Smishing else it is classified as legitimate message.

As shown in Fig. 1, a smishing SMS contains either a URL, a self answering link, a phone number or an email id in it. According to our study, without any of these items, an SMS is not expected to cause any malicious activity. Presence of malicious keywords may indicate a spam SMS or an advertising campaign but it cannot

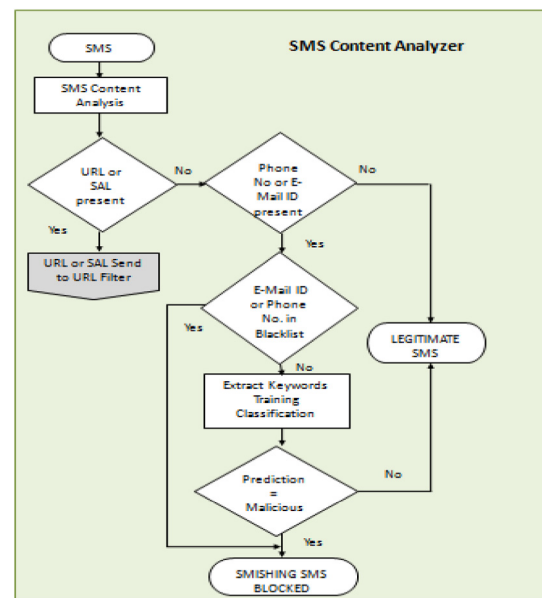


Fig. 4. Flowchart of SMS Content Analyzer.

be Smishing. So we are checking for the presence of smishing keywords only when there is the presence of phone number or email id in the SMS. On the other hand, if keywords are not malicious, it might be a phone number or email id forwarded by a known sender. So, it might be a legitimate SMS. Hence, we are trying to eliminate false positive results at every possible step. If a URL is present in the message, we further analyze the behavior of the URL to come to a conclusion. A sender in contacts list might forward a malicious link without knowing its maliciousness. So all the links forwarded by known contacts also need to be analyzed for maliciousness. So we are not considering the factor of 'sender in contacts list' for our SMS filtering. This helps in avoiding false negative results.

3.2.2. URL Filter

URL Filter analyzes the URL contained in the message as depicted in Fig. 5. The entire working scenario of URL Filter is drafted in the following steps:

Step 2(a): In URL Filter (Fig. 5), short URL is converted to long URL.

Step 2(b): Now, system check for the presence of URL in blacklist. If it does exist in the blacklist, then the message is categorized as a Smishing message.

Step 2(c): If the URL does not exist in the blacklist, URL Filter verifies four features of the URL, namely, age of domain, presence of @tag, presence of hyphen and number of dots present in the URL to check the authenticity of the URL.

Step 2(d): If the threshold of the above features is greater than or equal to three, then message is categorized as smishing else if the threshold is less than three, URL is handed over to Source Code Analyzer (Step-3(a)).

In URL Filter, we are checking for presence of URL in blacklist. URL blacklist is constructed from PhishTank [29], which is a website where users and researchers can share phishing data. But most of the URLs are not available in blacklists as attackers keep on changing the domains and blacklists are not updated frequently. So we are inspecting the authenticity of the URL by checking some malicious features of the URL and age of domain. Age of domain is a significant factor for detection of phishing sites as most of the phishing websites have an age of domain less than 6 months as they frequently makes new domains. As

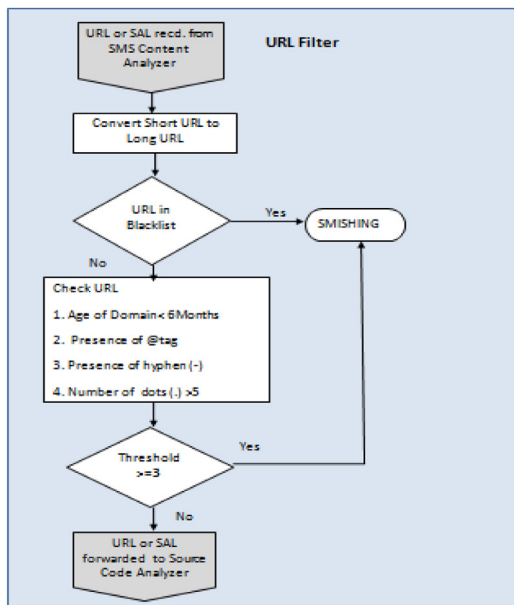


Fig. 5. Flowchart of URL Filter.

shown in Fig. 1, a malicious URL either prompts the user to fill user credentials or sensitive information like credit card details in a form provided or it downloads a malicious APK file which might turn out to be malware. So, once URLs are filtered by the URL Filter, Source Code Analyzer checks for the form tag in source code first to ensure that the attacker is not prompting the user to fill any form provided in the website. If form tag is not found, we check for the malicious file download activity by passing the URL to APK Download Detector.

3.2.3. Source Code Analyzer

Source Code Analyzer analyzes the source code of the URL without invoking the URL as depicted in Fig. 6. The entire working scenario of Source Code Analyzer is drafted in the following steps:

Step 3(a): In Source Code Analyzer (Fig. 6), source code of the URL is analyzed to check for the presence of form tag in it. If form tag is not present in the source code, URL is processed to APK Download Detector.

Step 3(b): If form tag is present in the source code, source code is further analyzed to check whether the domain name of request URL in the source code is different from the domain of the actual URL. If the domain is different, message is classified as Smishing.

Step 3(c): If the domain of the request URL is the same as the actual URL, URL is transferred to APK Download Detector (Step-4(a)) to further inspect the behavior of the URL.

Source Code Analyzer inspects the source code of the target URL without invoking the URL. This avoids any malicious activity by the phishing website until the analyzing process is completed and malicious URLs are identified. Form tag might be a login page of a legitimate shopping website which prompts the user to fill user credentials. So, the presence of a form tag alone cannot predict maliciousness. Hence, we are matching the URL domain in the source code with the actual URL domain to check its legitimacy which again helps us in reducing the false positive results. For most of the legitimate websites, the domain name is same in the source code.

3.2.4. APK Download Detector

The flowchart of APK Download Detector is depicted in Fig. 7. The entire working scenario of APK Download Detector is drafted in the following steps:

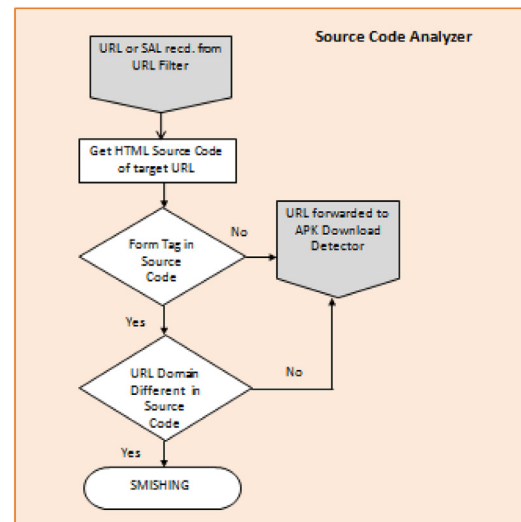


Fig. 6. Flowchart of Source Code Analyzer.

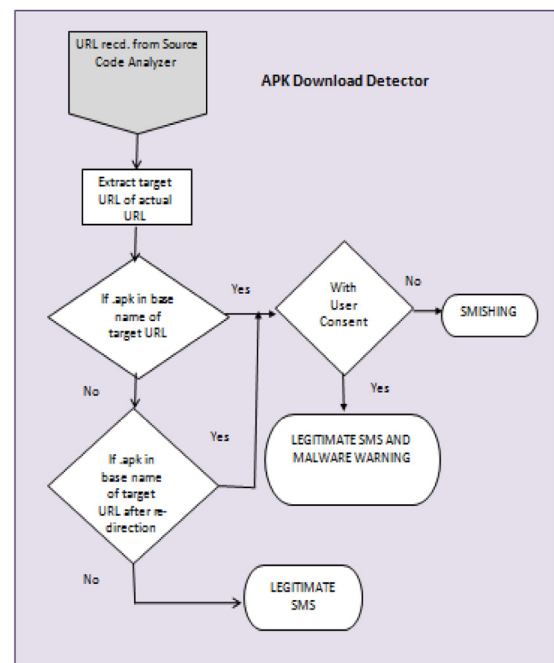


Fig. 7. Flowchart of APK Download Detector.

Step 4(a): Now the APK Download Detector (Fig. 7) extracts the target URL of the actual URL.

Step 4(b): If .apk extension is present in the target URL, the system checks whether user consent is taken while downloading the file. If user consent is not taken, the message is declared as a Smishing message. On the other hand, if user consent is taken, the message is regarded as a Legitimate message.

Step 4(c): If an APK is not downloaded in first access, then the system checks whether .apk extension is present in the target URL after re-direction of the page. If yes, then we go to step-4(b) to check for the user consent else if APK file is not downloaded after the re-direction of the page, the message is regarded as Legitimate message.

Apk Download Detector checks for any file download without invoking the URL to avoid any attack while accessing the URL. If the file is downloaded after taking user consent, it might be

Table 1
Popularity of features used in designing phishing websites.

Feature	Frequency in %
Long URL	47.2
Age of Domain	93.5
Presence of hyphen (-)	29.4
Number of dots in URL	10.2
Form tag in source code	96.5
Difference in domain	98.3
APK Download	56.4

an application of any legitimate website like Flipkart or Pinterest. Hence, if user consent is taken, we regard the message as legitimate. We are also checking for malicious file download after re-direction of the page.

3.3. Popularity of features used in the system

In case of a URL is detected in the message, we used some features of the phishing websites in order to detect the phishing attack. Some features commonly used by the attackers in designing phishing websites are identified. To find out which feature is most popular in designing phishing websites, we calculated the number of appearance of each feature in the phishing dataset [29]. The percentage of appearance i.e. frequency of each feature used in our system is shown in Table 1. The results showed that 'Difference in domain' is most popular feature in designing phishing websites. Form Tag, Age of Domain and APK download also appeared in most of the phishing websites with a frequency of 96.5%, 93.5% and 56.4% respectively. That is why we have tried to cater the features of higher percentage with utmost care.

4. Implementation results and evaluation

In this section, we describe the implementation details and evaluation results of the proposed system. A prototype of the Smishing Detector is developed using python in Jupyter Notebook. The system is developed into four parts, namely, SMS Content Analyzer, URL Filter, Source Code Analyzer and APK Download Detector. Further, these four modules are integrated to get a final prototype of the whole system. The system is finally evaluated using the dataset. Dataset is collected from the research work contributed by the author Almeida [30], a contribution to the study of SMS Spam Filtering. This dataset contains a total of 5574 messages in which 4827 are ham messages and 747 are spam messages. As per our research, smishing dataset is not publicly available until now. But smishing message is a spam message that strives to steal sensitive data from the user. So, we have extracted some smishing messages from spam dataset because smishing messages are part of spam messages. Also, we have extracted 284 smishing images from [pinterest.com](https://www.pinterest.com) [9]. We have added these messages into the dataset after converting it into text form. Our final dataset is a total of 5858 messages which contains 538 smishing messages and 5320 ham messages. Smishing detection is a binary classification problem in which either a message is smishing or it is legitimate.

Fig. 8 depicts that the model has four interfaces: SMS Content Analyzer, URL Filter, Source Code Analyzer and APK Download Detector. SMS Content Analyzer displays the phone number of the sender and the text message received. It provides three buttons for the convenience of the user. If the user is already aware of the genuineness of the message, then the user can tap the NEXT button to skip the verification of the particular message and go to next message or the user can tap the Go BACK button to go to the previous screen. On the other hand, if the user is tapping

Table 2
Performance of machine learning classifiers.

Classifiers	Precision	Recall	F1-Score	Accuracy
Naive Bayes Classifier	0.93	0.92	0.92	91.6
Random Forest classifier	0.88	0.82	0.83	82.3
Decision Tree classifier	0.91	0.88	0.89	88.2

on VERIFY button, the message is evaluated by the SMS Content Analyzer and the system displays the result and it redirects the user to URL Filter screen. URL Filter and Source Code Analyzer screen provide the user with the option of skipping that particular step by tapping the SKIP button as shown in Figs. 8(b) and 8(c). APK Download Detector module can be skipped by tapping on NEXT button which redirects the user to the next message. Hence, the user can navigate to next message either from the SMS Content Analyzer module or APK Download Detector module.

Fig. 9(a) shows a legitimate message identified by the proposed system. This interface provides three buttons: READ Button, NEXT Button and GO BACK Button. The message is recognized as genuine by the model, hence, the user is permitted to access the message by clicking the READ Button or user has the option to tap the NEXT button to go to the NEXT message. Go Back button navigates the user to the previous screen. Fig. 9(b) shows the smishing message recognized by the system. As the message is categorized as smishing by the model, the DELETE button deletes the messages without navigating to the inbox, and the NEXT button allows the user to move to the next message.

SMS Content Analyzer analyzes the text content of the SMS. Here, we check for the presence of email id, phone number and URL in the message. We segregated the messages based on its contents. Fig. 10 shows the result of the SMS content analysis. Out of 5858 messages, 5122 messages are segregated and declared as legitimate messages in this stage, as they do not contain any of the malicious contents like email id, phone number, or URL.

Text pre-processing is done on messages containing email id and phone number. Text pre-processing is conducted to convert the text into a form that can be used for text analysis. Text pre-processing includes removing all punctuations and special strings, converting each word to lower case, splitting words to tokenize, stemming i.e. converting each word to its root form and preparing a word vector corpus. Fig. 11 shows the results of the text pre-processing module.

After text pre-processing, we categorize the messages on the basis of keywords included in it. Keywords contained in the message are classified using TfidfVectorizer and machine learning classifiers. TfidfVectorizer converts each word in the message to feature index in the feature vector matrix which is used as input to machine learning classifiers. Machine learning classifier predicts the result based on the learning from the feature vector matrix. We have used three machine learning algorithms for the classification and comparison purpose, namely, Naive Bayes, Random Forest, and Decision Tree. We have evaluated our dataset for keyword classification using these three algorithms and we are getting the best results using Naive Bayes Algorithm for the particular dataset.

The model evaluated the performance of three well-known machine learning classifiers as depicted in Table 2. Fig. 12 show the performance of machine learning algorithms. In this Figure, the x-axis displays the machine learning algorithms, and the y-axis displays the performance of the algorithms. For the implemented dataset, Naive Bayes gives the best performance and Random Forest gives the least performance.

Source Code Analyzer analyzes the presence of any form tag in the source code of the URL. It fetches the actual URL of the short URL provided. Then it accesses the source code of the



(a): Interface of SMS Content Analyzer;



(b): Interface of URL Filter

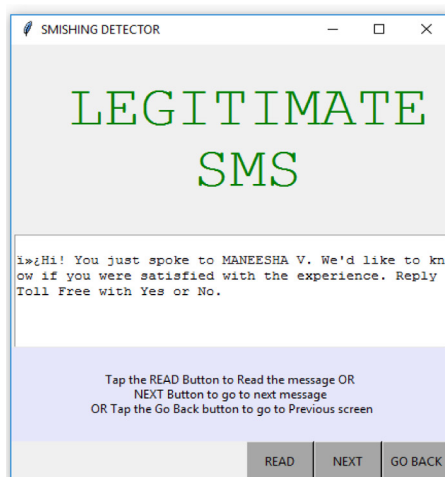


(c): Interface of Source Code Analyzer;

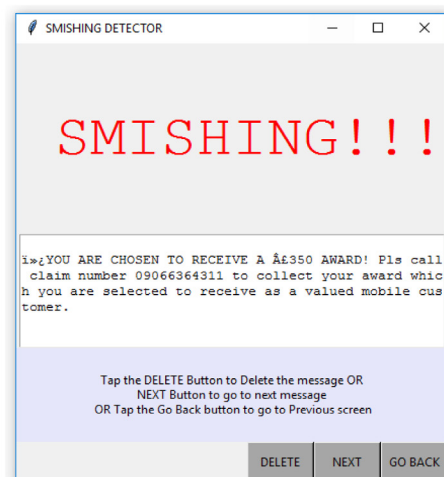


(d): Interface of APK Download Detector

Fig. 8. User interfaces of Smishing Detector.



(a): Legitimate message detected by the system;



(b): Smishing message detected by the system

Fig. 9. Results displayed by Smishing Detector.


```

["ham\tYa, i'm referin to mei's ex wat... No ah, waitin 4 u to treat, somebody shld b rich liao...So gd, den u dun
have to work frm tmr onwards...\n"]
['ham\tMiles and smiles r made frm same letters but do u know d difference..? smile on ur face keeps me happy even
though I am miles away from u.. :-)keep smiling.. Good nyt\n']
["ham\tBy the way, i've put a skip right outside the front of the house so you can see which house it is. Just pull
up before it.\n"]
['ham\tCan you pls send me that company name. In saibaba colany\n']
['ham\tNo. I dont want to hear anything\n']
['ham\tYou are a big chic. Common. Declare\n']
['ham\tThats cool. I want to please you...\n']
['ham\tGoing to join tomorrow.\n']
['ham\tI want to tell you how bad I feel that basically the only times I text you lately are when I need drugs\n']
['ham\tTotal disappointment, when I texted you was the craziest shit got :(\n']
['ham\tIts just the effect of irritation. Just ignore it\n']
['ham\tWhat about this one then.\n']
["ham\tI think that tantrum's finished so yeah I'll be by at some point\n"]
['ham\tCompliments to you. Was away from the system. How your side.\n']
['ham\tHappened here while you were adventuring\n']
['ham\tHey chief, can you give me a bell when you get this. Need to talk to you about this royal visit on the 1st j

```

Fig. 10. Results of text messages segregation based on its contents.

```

After Regular Expression - Message 0 : Go until jurong point crazy Available only in bugis n great world la e
buffet Cine there got amore wat

Lower case Message 0 : go until jurong point crazy available only in bugis n great world la e buffet cine t
here got amore wat

After Splitting - Message 0 : ['go', 'until', 'jurong', 'point', 'crazy', 'available', 'only', 'in', 'bugis',
'n', 'great', 'world', 'la', 'e', 'buffet', 'cine', 'there', 'got', 'amore', 'wat']

After Stemming - Message 0 : ['go', 'jurong', 'point', 'crazi', 'avail', 'bugi', 'n', 'great', 'world', 'la',
'e', 'buffet', 'cine', 'got', 'amor', 'wat']

Final Prepared - Message 0 : go jurong point crazi avail bugi n great world la e buffet cine got amor wat

```

Fig. 11. Results of SMS text pre-processing done on messages containing email id and phone number.

actual URL without actually invoking the URL. Now, the system search for a form tag in the whole source code. The result is displayed according to the search outcome obtained. The result of the procedure is depicted in Fig. 13.

APK Download Detector checks for any file download without invoking the URL. Fig. 14 shows the whole procedure of performing an APK Download Check. The first line shows the URL which is in short URL. The second line shows the result of fetching the actual link address from the short URL. The third line shows extracting the file name of the file downloaded from the link without taking user consent. If the file is downloaded without user consent, the link is declared as malicious and hence the message is categorized as SMISHING.

We have evaluated our dataset with 5-fold cross-validation where we partition the data into 5 equally sized subsets or folds. In the first iteration, first subset is used to test the model and rest 4 subsets are used to train the model. In the second iteration, 2nd subset is used as the testing set while the rest serve as the training set. This process is repeated until each subset of the 5 folds has been used as the testing set. 5-fold cross-validation uses

80% of data for training purpose, and remaining 20% data for testing purpose. After 5 iterations, our system gave a final accuracy of 96.29%. Cross-validation results are depicted in Table 3.

The result of the prototype evaluated and experimented shows that the model gives an accuracy of 91.6% using Naive Bayes Classifier in SMS Content Analyzer module. Further, the model shows a final accuracy of 96.29% after evaluating all four modules. Hence, through our model evaluation, it is concluded that this system is efficient in detecting smishing messages received in mobile devices and thereby protecting the users from probable threats.

As we have followed a flow based approach consisting of a set of diverse rules, the attacker needs to breach each rules in a module to reach the next stage in a particular module. We have four modules in our system, it is almost impossible for the attacker to predict the whole flow of the system, to circumvent each of the diverse rules applied in each module and to reach to the final decision stage. Hence, this model is a proof against the attacker who is constantly trying to invent different techniques to circumvent the smishing detection system.

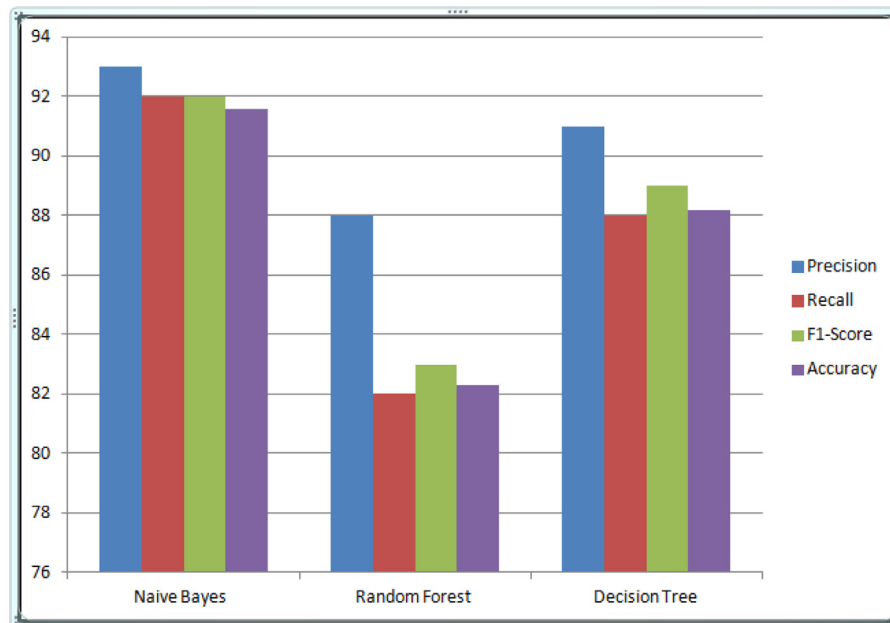


Fig. 12. Performance of Machine learning algorithms for SMS keyword classification.

```

URL is https://bit.ly/1ezaz2m
response is https://f-droid.org/FDroid.apk
No Form tag found

URL is https://play.google.com/store/apps/details?id=net.runserver.tripeaks
response is https://play.google.com/store/apps/details?id=net.runserver.tripeaks
Form Tag is found

URL is https://play.google.com/store/apps/details?id=com.extremeenjoy.news.kannadacinema
response is https://play.google.com/store/apps/details?id=com.extremeenjoy.news.kannadacinema
Form Tag is found

URL is https://play.google.com/store/apps/details?id=com.extremeenjoy.news.anweshanam
response is https://play.google.com/store/apps/details?id=com.extremeenjoy.news.anweshanam
Form Tag is found

```

Fig. 13. Results of Form Tag check in the HTML Source code by Source Code Analyzer.

```

URL is https://bit.ly/1ezaz2m
response is https://f-droid.org/FDroid.apk
File name is: FDroid.apk
THE URL IS MALICIOUS
*****

URL is https://play.google.com/store/apps/details?id=net.runserver.tripeaks
response is https://play.google.com/store/apps/details?id=net.runserver.tripeaks
File name is: details?id=net.runserver.tripeaks
THE URL LOOKS CLEAN
*****

URL is https://play.google.com/store/apps/details?id=com.extremeenjoy.news.kannadacinema
response is https://play.google.com/store/apps/details?id=com.extremeenjoy.news.kannadacinema
File name is: details?id=com.extremeenjoy.news.kannadacinema
THE URL LOOKS CLEAN
*****

```

Fig. 14. URL Prediction on the basis of APK Downloading.

Table 3

Cross-validation results of the Proposed Approach.

Iterations	Data					Accuracy
Iteration 1	Subset 1	Subset 2	Subset 3	Subset 4	Subset 5	96.42
Iteration 2	Subset 1	Subset 2	Subset 3	Subset 4	Subset 5	97.85
Iteration 3	Subset 1	Subset 2	Subset 3	Subset 4	Subset 5	96.41
Iteration 4	Subset 1	Subset 2	Subset 3	Subset 4	Subset 5	94.38
Iteration 5	Subset 1	Subset 2	Subset 3	Subset 4	Subset 5	96.40
Total						96.29

Table 4

Comparison of our proposed system with the existing researches.

Security Requirements	S-Detector [13]	Rule-Based [14]	SmiDCA [11]	Feature-Based [17]	Proposed System
Approaches Used	Rule-Based Flowchart	Rule-Based	Heuristic-based Feature selection	Machine learning classification of SMS Features	Rule-Based Flowchart
Classification Approach	Naive Bayes	Decision Tree, RIPPER and PRISM	Random Forest, Decision Tree, AdaBoost and SVM	Logistic Regression, Neural Network, Random Forest, Naive Bayes and SVM	Naive Bayes
Dataset Used	Not Specified	Almeida [30]-spam dataset manually filtered to make a new smishing dataset of 5169 messages of which 362 are smishing messages and 4807 are ham.	Almeida [30]-spam dataset of 5574 messages of which 747 are spam messages and 4827 are ham.	Almeida [30]-spam dataset manually filtered to make a new smishing dataset of 5169 messages of which 362 are smishing messages and 4808 are ham.	Almeida [30]-spam dataset manually filtered to make a new smishing dataset of 5858 messages of which 538 are smishing messages and 5320 are ham.
Keywords Classification	✓	✓	✓	✓	✓
presence of URL	✓	✓	✓	✓	✓
Presence of Phone no. and Email id in the message	✓	✓	✓	✓	✓
Phone no. and Email id in blacklist	X	X	X	X	✓
URL in Blacklist	X	X	X	X	✓
Check for login page	X	X	X	X	✓
Difference in URL Domain	X	X	X	X	✓
APK Download	✓	X	X	X	✓
APK Download after re-direction	✓	X	X	X	✓
User consent while downloading APK	X	X	X	X	✓

5. Comparative analysis

We analyzed the existing researches related to Smishing and compared the related works based on the security measures. Table 4 represents a comparative analysis of recent researches with the proposed system for Smishing detection systems.

The existing Smishing detection systems proposed by other authors and the proposed system commonly check for the presence of URL when an SMS is received, and they also do the content analysis of the text message received. But Content Analysis alone cannot determine the maliciousness of a message until and unless we further check the maliciousness of the URL involved.

Verification for the presence of self answering link, phone number and email id is not done in Smishing detection system proposed by Joo et al. [13]. They have analyzed the URL for download of any APK file but have not checked for the presence

of any login page which prompts the user to fill user credentials. Ankit et al. [14] have followed a rule-based smishing detection system considering the contents of the message but they have not analyzed the behavior of the URL. SmiDCA proposed by Sonowal et al. [11] has presented a heuristics-based system considering the contents of an SMS and the keywords present in it. This system does not check for the maliciousness of the URL, whether it downloads a malicious file or prompts the user to fill user credentials in a form provided. As per our study, Smishing is a threat in which an attacker sends an SMS to the user, and that SMS includes links to user interfaces, malicious applications, web pages that prompt user to enter their credentials. Hence, we cannot classify a message as smishing if the link contained in it is legitimate even if the message contains malicious keywords. Hence, we strongly sense the need for checking the behavior of the URL further to analyze the maliciousness of the link.

Smishing Classifier proposed by Goel et al. [10] has done similar work but they have not eliminated the false-positive results. If an APK is downloaded, they have classified it a smishing but an APK can be the legitimate app of a shopping website like Flipkart. Similarly, login page and self-answering link can be legitimate too in case of a message received from Facebook or Amazon asking the user to log in to their website for availing some offers. Hence, we need to further check the legitimacy of the website to avoid false-positive results. If a URL or Self Answering Link is not present in a message, keywords threshold alone cannot determine the maliciousness because as per our study, a message without any link, mobile number or E-mail ID cannot cause any malicious activity even if it has high threshold keywords present in it. Verification for the User Consent while downloading APK file performed in the proposed system is not done by any of the authors till now. It considerably helps to detect the malicious file downloading on a mobile device. This check also helps to reduce false-positive results.

6. Conclusion and future work

Smishing is a critical attack involved with mobile devices which is rising in this mobile era. Hence, this paper proposed an efficient model titled Smishing Detector to detect and block Smishing attacks. The proposed model is using SMS content analysis and URL inspection method to classify smishing messages from legitimate messages. SMS Content Analyzer is the module to analyze the contents of the message. URL Filter, Source Code Analyzer and APK Download Detector are the modules to inspect the behavior of URL contained in the message. We have also developed a prototype of the system using the tkinter package. This prototype provides user-friendly buttons to skip some of the steps involved in the system. Machine learning algorithms are used to classify messages on the basis of smishing keywords. Naive Bayes classifier shows the best accuracy for the keyword classification in our proposed system for the particular dataset involved. After integrating all the four modules, the final prototype experimented shown an accuracy of 96.29%.

A comparison of our model with existing models displayed that this system comprises more security requirements as compared to other proposed models. This system is expected to deliver more effective security against attacks in terms of identifying smishing messages and preventing false-positive results. The practical implementation of this work can be integrated with the Android platform and can be used as an application to detect the smishing messages. This application can be used to identify smishing message when a message is received and the message can be discarded or saved based on the result.

In future, we are planning to incorporate more techniques to the proposed model in order to prevent more intelligent and versatile threat methods. This system is lacking security in ensuring the genuineness of the application downloaded in APK Download Detector module. Hence, to ensure application security, we are also planning to embed Malware detector with APK Download Detector to identify malicious apps in our future work. This will focus on more research work to provide security against personal information leakage and to detect malicious applications downloaded.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Sandhya Mishra: Conceptualization, Methodology, Software, Data curation, Writing - original draft, Visualization, Investigation. **Devpriya Soni:** Conceptualization, Supervision, Validation, Writing - review & editing.

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